

Test 4.2 Practice Problems.

1. Mark each series as Absolutely convergent, Conditionally Convergent, or Divergent.

(a) $\sum_{n=1}^{\infty} \frac{3}{n^2}$ *Abs. Converge*

(b) $\sum_{n=1}^{\infty} \frac{2^n}{4^{n+3}}$ *Absolutely Converges*

(c) $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2+1}$ *Conditionally Convergent*
Converges (by AST)
 $\sum \frac{n}{n^2+1}$ *Diverges*

(d) $\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{n+5n^2}$ *Diverges*

2. Use the ratio test to determine if these series converge or diverge.

(a) $\sum_{n=1}^{\infty} \frac{5^n}{n^3 \cdot 3^n}$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

(b) $\sum_{n=1}^{\infty} (-1)^n \frac{n^9 \cdot 7^n}{n!}$ $\lim_{n \rightarrow \infty} \frac{(n+1)^9 7^{n+1}}{(n+1)!} \cdot \frac{n!}{n^9 7^n} = \lim_{n \rightarrow \infty} \frac{(n+1)^9}{n^9} \cdot \frac{7^{n+1}}{7^n} \cdot \frac{n!}{(n+1)!} = 0 < 1$
 $= \lim_{n \rightarrow \infty} \underbrace{\frac{(n+1)^9}{n^9}}_1 \cdot \underbrace{\frac{7^{n+1}}{7^n}}_7 \cdot \underbrace{\frac{n!}{(n+1)n!}}_0 = 0 < 1$ C

$$(c) \sum_{n=1}^{\infty} \frac{n^3 \ln n}{2^n} \quad \lim_{n \rightarrow \infty} \frac{(n+1)^3 \ln(n+1)}{2^{n+1}} \cdot \frac{2^n}{n^3 \ln n}$$

$$= \lim_{n \rightarrow \infty} \underbrace{\frac{(n+1)^3}{n^3}}_1 \cdot \underbrace{\frac{\ln(n+1)}{\ln(n)}}_1 \cdot \underbrace{\frac{2^n}{2^{n+1}}}_{\frac{1}{2}} = \frac{1}{2} < 1 \therefore \boxed{\text{Converge}}$$

$$(d) \sum_{n=1}^{\infty} \frac{2^{n^2}}{n!} \quad \lim_{n \rightarrow \infty} \frac{2^{(n+1)^2}}{(n+1)!} \cdot \frac{n!}{2^{n^2}} = \lim_{n \rightarrow \infty} \frac{2^{n^2+2n+1}}{2^{n^2}} \cdot \frac{n!}{(n+1)n!} = \lim_{n \rightarrow \infty} \frac{2^{2n+1}}{n+1} = \infty > 1 \therefore D$$

3. Find the interval of convergence for the power series.

$$(a) \sum_{n=1}^{\infty} (-1)^n \frac{(x-9)^n}{3^n} \quad \lim_{n \rightarrow \infty} \frac{(x-9)^{n+1}}{3^{n+1}} \cdot \frac{3^n}{(x-9)^n} = \lim_{n \rightarrow \infty} \frac{(x-9)^{n+1}}{(x-9)^n} \cdot \frac{3^n}{3^{n+1}} = \frac{(x-9)}{3}$$

$$\underline{x=6}$$

$$\sum (-1)^n \frac{(6-9)^n}{3^n} = \sum (-1)^n (-1)^n = \sum 1 \quad D$$

$$-1 < \frac{x-9}{3} < 1$$

$$-3 < x-9 < 3$$

$$6 < x < 12$$

$$\underline{x=12}$$

$$\sum (-1)^n \frac{(12-9)^n}{3^n} = \sum (-1)^n \quad D$$

$$\boxed{(6, 12)}$$

$$(b) \sum_{n=1}^{\infty} \frac{x^{n+2}}{n^2 \cdot 2^n} \quad \lim_{n \rightarrow \infty} \frac{x^{n+2}}{(n+1)^2 \cdot 2^{n+1}} \cdot \frac{n^2 \cdot 2^n}{x^{n+1}} = \lim_{n \rightarrow \infty} \underbrace{\frac{x^{n+2}}{x^{n+1}}}_x \cdot \underbrace{\frac{n^2}{(n+1)^2}}_1 \cdot \underbrace{\frac{2^n}{2^{n+1}}}_{\frac{1}{2}} = \frac{x}{2}$$

$$\underline{x=2} \quad \sum \frac{2^{n+1}}{n^2 \cdot 2^n} = \sum \frac{2}{n^2} \quad C$$

$$-1 < \frac{x}{2} < 1 \\ -2 < x < 2$$

$$(c) \sum_{n=1}^{\infty} (-1)^n \frac{(2x+4)^n}{4n+1}$$

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (2)^{n+1}}{n^2 \cdot 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 2}{n^2}$$

$$[-2, 2]$$

4. Write out each series using sigma notation.

(a) $x^2 + 3x^6 + 9x^{10} + 27x^{14} + \dots$

$\sum_{n=0}^{\infty} 3^n x^{4n+2}$

difference in powers
first power

(b) $\frac{x}{6} + \frac{x^4}{12} + \frac{x^7}{24} + \frac{x^{10}}{48} + \frac{x^{13}}{96} + \dots$

$\sum_{n=0}^{\infty} \frac{x^{3n+1}}{6 \cdot 2^n}$

$\sum_{n=0}^{\infty} \frac{x^{3n+1}}{6 \cdot 2^n}$

(c) $\frac{x^7}{5} + \frac{4x^{17}}{25} + \frac{16x^{27}}{125} + \frac{64x^{37}}{625} + \frac{256x^{47}}{3125} + \frac{1024x^{57}}{15625} + \dots$

$\sum_{n=0}^{\infty} \frac{x^{10n+7} \cdot 4^n}{5 \cdot 5^n} = \sum_{n=0}^{\infty} \frac{x^{10n+7} \cdot 4^n}{5^{n+1}}$

5. Write out the first four nonzero terms in the power series representation of each function:

(a) $g(x) = \frac{x^3}{1-x^2}$

$q = x^3$
 $r = x^2$

$x^3 + x^5 + x^7 + x^9 + \dots$

$\sum_{n=0}^{\infty} x^{2n+3}$

(b) $g(x) = \frac{1}{5-2x} \cdot \frac{1/5}{1/5} = \frac{1/5}{1-\frac{2}{5}x} = \frac{1}{5} + \frac{2}{25}x + \frac{4}{125}x^2 + \frac{8}{625}x^3 + \dots$

$\sum_{n=0}^{\infty} \frac{1}{5} \left(\frac{2}{5}x\right)^n = \sum_{n=0}^{\infty} \frac{2^n x^n}{5^{n+1}}$

(c) $g(x) = \frac{8x^3}{2+x^5} \cdot \frac{1/2}{1/2} = \frac{4x^3}{1-(-\frac{1}{2}x^5)} = 4x^3 - 2x^8 + 1x^{13} - \frac{1}{2}x^{18} + \frac{1}{4}x^{23} + \dots$

$= \sum_{n=0}^{\infty} ar^n = \sum_{n=0}^{\infty} 4x^3 \left(-\frac{1}{2}x^5\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n x^{5n+3}}{2^{n-2}}$

6. Find a series for each integral. Write out at least 4 nonzero terms.

$$\begin{aligned}
 \text{(a)} \quad \int e^{x^3} dx &= \int 1 + \frac{(x^3)}{1!} + \frac{(x^3)^2}{2!} + \frac{(x^3)^3}{3!} + \dots dx \\
 &= \int 1 + x^3 + \frac{1}{2} x^6 + \frac{1}{6} x^9 + \dots dx \\
 &= K + x + \frac{1}{4} x^4 + \frac{1}{14} x^7 + \frac{1}{60} x^{10} + \dots
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \int \frac{\sin(x^2)}{x} dx & \quad \frac{\sin(x^2)}{x} = \frac{(x^2)}{x} - \frac{(x^2)^3}{3!x} + \frac{(x^2)^5}{5!x} - \frac{(x^2)^7}{7!x} + \dots \\
 & \int x - \frac{1}{6} x^5 + \frac{1}{120} x^9 - \frac{1}{5040} x^{13} + \dots dx \\
 & = K + \frac{1}{2} x^2 - \frac{1}{36} x^6 + \frac{1}{1200} x^{10} - \frac{1}{70560} x^{14} + \dots
 \end{aligned}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Answers

1. (a) AC (b) AC (c) CC (d) D

2. (a) D (b) C (c) C (d) D

3. (a) $(6, 12)$ (b) $[-2, 2]$ (c) $(-\frac{5}{2}, -\frac{3}{2}]$

4. (a) $\sum_{n=0}^{\infty} 3^n x^{4n+2}$

(b) $\sum_{n=0}^{\infty} \frac{x^{3n+1}}{6 \cdot 2^n}$

(c) $\sum_{n=0}^{\infty} \frac{4^n x^{10n+7}}{5^{n+1}}$

5. (a) $x^3 + x^5 + x^7 + x^9 + \dots$

(b) $\frac{1}{5} + \frac{2}{25}x + \frac{4}{125}x^2 + \frac{8}{625}x^3 + \dots$

(c) $4x^3 - 2x^8 + x^{13} - \frac{x^{18}}{2} + \frac{x^{23}}{4} \dots$

6. (a) $K + x + \frac{x^4}{4} + \frac{x^7}{14} + \frac{x^{10}}{60} + \frac{x^{13}}{312} + \dots$

(b) $K + \frac{x^2}{2} - \frac{x^6}{36} + \frac{x^{10}}{1200} - \frac{x^{14}}{70560} + \dots$